Evaluating Expressions with Radicals

Question Bank 4.05

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

For questions 1 through 6, evaluate.

1. $\sqrt{81}$

2. $\sqrt[3]{125}$

3. $\sqrt[3]{-8}$

4. $\sqrt{\frac{4}{25}}$

5. $\sqrt{5^2+11}$

6. $\sqrt[3]{3^3+37}$

For questions 7 and 8, evaluate. Each expression has a power and a radical.

7. Evaluate each expression.

a. $\sqrt{100} \cdot 2^{-2}$

b. $\left(-\frac{1}{2}\right)^2 + \sqrt{2^3 + 8}$

c. $\sqrt[3]{-64} - 3^{-1}$

8. Evaluate each expression.

a. $\sqrt[3]{27} - 1.4\left(\sqrt{3^2 - 5}\right)$

b. $\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$

c. $\sqrt{4^3 - 39} \cdot 5^{-3}$

Working backwards.

9. Find the value of n that makes each statement true. (The radicand is the unknown here, so it is not one of the listed perfect values until you find it.)

a. $\sqrt{n} = 12$

b. $\sqrt[3]{n} = -5$

c. $\sqrt{n+7} = 9$

Reading someone else's work.

10. Tonya writes $\sqrt{16+9} = \sqrt{16} + \sqrt{9} = 7$. Evaluate the expression correctly, and explain in one sentence what the radical bar does that her method ignores.

11. Which is the value of $\sqrt{36+64}$?

- ○ A. 14
- ○ B. 100
- ○ C. 10
- ○ D. 50

12. Select ALL of the expressions whose value is 3.

- ☐ A. $\sqrt{3^2+16}$
- ☐ B. $\sqrt[3]{2^2+23}$
- ☐ C. $\sqrt{4^2-7}$
- ☐ D. $\sqrt[3]{27}$
- ☐ E. $\sqrt[3]{-27}$
- ☐ F. $\sqrt{9}$